

Bellabeat Customer Trends

Project Owner: Sayed Ali Shahin	Date: May 15, 2026
Dataset: FitBit Tracker Data	Status: Production - Ready

Project Changelog

A complete overview of data cleanings, dropped variables, and required query rules.

Table / Source	Issue & Cleaning Action	Mandatory Analysis Rule
METs_minute_narrow	Identified 50 physiological extreme outliers (METs > 100). Checked for duplicates/nulls (0 found).	WHERE METs <= 100
daily_activity	verified as sub-daily (minute/hourly rows).	Treat rows as sub-daily metrics
daily_calories	Identified anomalous '0' calorie entries caused by device non-wear periods.	WHERE Calories > 0
hourly_calories	Passed checks completely (0 duplicates, 0 nulls). All metrics are normal.	Ready; no filters needed
intensity_hourly	Checks passed completely. Maximum value 180 verified as accurate (60 min x 3).	Ready; no filters needed
steps_daily	Identified anomalous '0' step data points caused by tracker non-wear days.	WHERE StepTotal > 0
sleep_day	Identified and isolated 3 precise pairs of duplicate data records.	SELECT DISTINCT mandatory
Sleep Minute Table	Purged 543 invalid duplicate records that contained critical missing NULL metrics.	Use cleaned output table
Hourly Steps Table	Baseline checks passed completely. Composite keys (Id, Date, Time) verified unique.	Ready; no filters needed
weight_fat_log_info	Verified completeness. Fat tracking column is 100% full. Weight and BMI spreads are valid.	Ready; no filters needed
Structural Schema Changes		

Data Dictionary

Field / Column	Standard Type	Data Unit	Description / Analytical Constraint
Id	INT	ID Code	Unique fitness tracker key matching distinct users across all logs.
Date / SleepDay	DATE	YYYY-MM-DD	Tracking calendar day of recorded user metrics.
Time	TIME	HH:MM:SS	Localized minute or hourly timestamp of registered activity records.
METs	INTEGER	Ratio	Metabolic Equivalent of Task exertion score. Filter with: WHERE METs <= 100.
Calories (Sub-Daily)	FLOAT	kcal	Energy burned at minute/hourly increments inside 'daily_activity' table.
Calories (Daily Summary)	INTEGER	kcal	Total kilocalories burned over 24 hours. Filter with: WHERE Calories > 0.
Intensity	INTEGER	Score (0-180)	Tracker vector score measuring hourly motion. Hourly capacity is 180.
StepTotal	INTEGER	Steps	Total steps taken within a full 24-hour window. Filter with: WHERE StepTotal > 0.
TotalMinutesAsleep	INTEGER	Minutes	Total minutes of asleep runtime tracked. (Short logs signify naps).
TotalTimeIn Bed	INTEGER	Minutes	Total horizontal bed duration. Difference yields sleep latency numbers.
WeightKg	FLOAT	Kilograms	User body weight logs. Spans values between 52kg and 133kg.
Fat	FLOAT	Percentage	Body fat ratio logging metrics. Complete data profile without missing parameters.
BMI	FLOAT	Index Score	Body Mass Index metric calculating weight-to-height parameters (21-47 range).